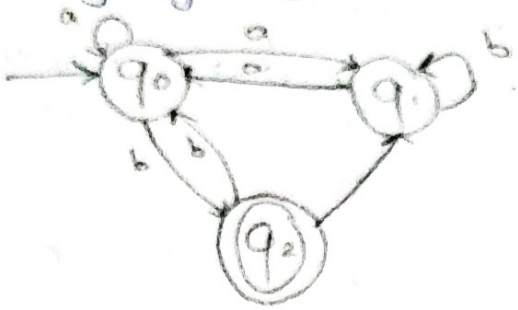
8. Consider the NFA given below which accepts some language L over the alphabet $\Sigma = \{a, b\}$. Construct DFA that represents the same language L.



Step 1: Identify the NFA

From the given diagram:

- Initial state: q_0
- Final state: q_2
- Alphabet: $\Sigma = a, b$

The transitions are:

State	Input a	Input b
q_0	q_0, q_1	q_2
q_1	q_0	q_1, q_2
q_2	—	q_0, q_2

(b) Step 2: Apply the Subset Construction Method

In the subset construction method, each DFA state represents a set of NFA states.

Initial DFA state:

$A = \{q_0\}$

1) From $A = q_0$
   - On input a: $q_0, q_1 = B$
   - On input b: $q_2 = C$

2) From $B = q_0, q_1$
   - On input a: $q_0, q_1 = B$
   - On input b: $q_1, q_2 = D$

3) From $C = q_2$
   - On input a: $\emptyset = E$
   - On input b: $q_0, q_1 = B$

4) From $D = q_1, q_2$
   - On input a: $q_0 = A$
   - On input b: $q_0, q_1, q_2 = F$

5) From $E = \emptyset$
   - On input a: $\emptyset = E$
   - On input b: $\emptyset = E$

e) From F: $q_0 \cdot q_1 \cdot q_2$
- On input a: $q_0 \cdot q_1 = B$
- On input b: $q_0 \cdot q_1 \cdot q_2 = F$

Step 3: DFA Transition Table

DFA State	NFA State Set	Input a	Input b
→ A	q_0	B	C
B	$q_0 \cdot q_1$	B	D
*C	q_2	E	B
D	$q_1 \cdot q_2$	A	F
E	$\emptyset$	E	E
*F	$q_0 \cdot q_1 \cdot q_2$	B	F

Initial state: A
Final states: C, D, F
A DFA state is final if its set contains the NFA's final state q_2.

2) Let L be the language of words of length atleast 2 whose last two letters are different from each other. For example abaab is in L but not bb, Construct an NFA for L. Then draw the equivalent DFA.

Step 1: Understand the Language

The alphabet is:

$$\Sigma = \{a, b\}$$

The conditions are:

1) The string must have a length of atleast 2.
2) The last two symbols must be different.

Therefore, a string is accepted if it ends with either:
- ab
- ba

Examples:

String	Last two symbols	Result
ab	ab	Accepted
ba	ba	Accepted
aab	ab	Accepted
bba	ba	Accepted
aa	aa	Rejected

bb	bb	Rejected
abab	ab	Accepted
abba	ba	Accepted

Part A: Construct the NFA

Step 2: Define the States

Use three states

- q_0: Initial state, reads any number of symbols.
- q_1: Guesses that the current symbol is the second-last symbol and reads a.
- q_2: Final state, confirms that the last two symbol are different.

Step3: Draw the NFA

The transitions are:

- $q_0 \xrightarrow{a,b} q_0$
- $q_0 \xrightarrow{a} q_1$
- $q_1 \xrightarrow{b} q_2$

(10) Also, to accept strings ending in ba, add:

- $q_0 \xrightarrow{b} q_1$
- $q_1 \xrightarrow{a} q_2$

Here, q_2 is the final state.

Step 4: NFA Transition Table

State	Input a	Input b
$\rightarrow q_0$	q_0, q_1	q_0, q_1
q_1	q_2	q_2
$*q_2$	$\emptyset$	$\emptyset$

This table isn't correct for the intended NFA. because q_1 must remember whether it guessed a or b. Use the corrected NFA below:

Use four states: q_0 q_a, q_b, q_f

Correct NFA Table

State	Input a	Input b
$\rightarrow q_0$	q_0, q_a	q_0, q_b
q_a	$\emptyset$	q_f
q_b	q_f	$\emptyset$
$*q_f$	$\emptyset$	$\emptyset$

Here:

- q_0 loops on both a and b.
- $q_0 \xrightarrow{a} q_a$, followed by $q_a \xrightarrow{b} q_f$, recognizes ab.
- $q_0 \xrightarrow{b} q_b$, followed by $q_b \xrightarrow{a} q_f$, recognizes ba.
- q_f is the final state.

Part B: Convert the NFA into an Equivalent DFA

Step 5: Apply Subset Construction

Let the DFA states be:

A = { q_0 }

B = { q_0, q_a }

C = { q_0, q_b }

D = { q_0, q_a, q_f }

E = { q_0, q_b, q_f }

F = { q_0, q_a, q_b }

G = { q_0, q_a, q_b, q_f }

Step 6: DFA Transition Table

DFA state	Input a	Input b
→ A	B	C
B	B	D
C	E	C
*D	B	G
*E	G	C
F	D	E
*G	G	G

Initial state: A

Final state: D, E, G

A DFA state is final if the corresponding set contains q_f.